

PROJECT PROPOSAL

CLASSROOM ALLOCATION MANAGEMENT SYSTEM

A Web-Based Application Using PHP and MySQL

Group Members

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This project proposal outlines the design and development of a centralized Classroom Allocation Management System aimed at improving scheduling accuracy, administrative efficiency, and optimal utilization of classroom resources through automation and structured data management.

1. Introduction

Classroom allocation is a crucial administrative task in educational institutions. Universities conduct multiple academic activities each day, including lectures, practical sessions, examinations, seminars, and workshops. Efficient allocation of classrooms ensures uninterrupted academic delivery and effective utilization of physical resources.

Manual classroom scheduling methods are prone to errors such as overlapping bookings, misallocation of room capacity, and inefficient space usage. These challenges increase administrative workload and reduce institutional efficiency. The proposed system introduces a centralized, web-based approach to automate classroom scheduling and minimize such issues.

2. Background of the Study

Advancements in information technology have transformed administrative operations across organizations. Educational institutions increasingly adopt web-based systems to manage academic scheduling and physical resources. Such systems provide centralized access, real-time updates, and improved coordination among departments.

A classroom allocation system ensures that available infrastructure is used effectively while reducing human error. Implementing such a system supports better planning, transparency, and long-term institutional growth.

3. Problem Statement

The current classroom allocation process faces challenges such as manual scheduling, lack of real-time availability tracking, frequent timetable conflicts, and inefficient use of classroom capacity. These issues become more severe as academic activities increase in scale and complexity.

4. Objectives of the Project

Main Objective:

To develop a web-based Classroom Allocation Management System that automates and optimizes classroom scheduling.

Specific Objectives:

- Centralize classroom and scheduling data
- Reduce timetable conflicts
- Improve classroom utilization
- Support role-based access
- Generate administrative reports

5. Scope of the Project

The system will manage classroom details, scheduling of lectures and examinations, availability checking, and report generation. The system excludes online learning, attendance tracking, and grading functionalities.

6. Proposed System Description

The proposed system will be a secure, browser-based application featuring authentication, conflict detection, and centralized database management using MySQL. The system will reduce dependency on manual scheduling and improve operational accuracy.

7. System Benefits

The system will reduce administrative workload, prevent scheduling conflicts, enhance decision-making, and ensure optimal use of classroom infrastructure.

8. Technologies Used

Frontend: HTML, CSS, JavaScript

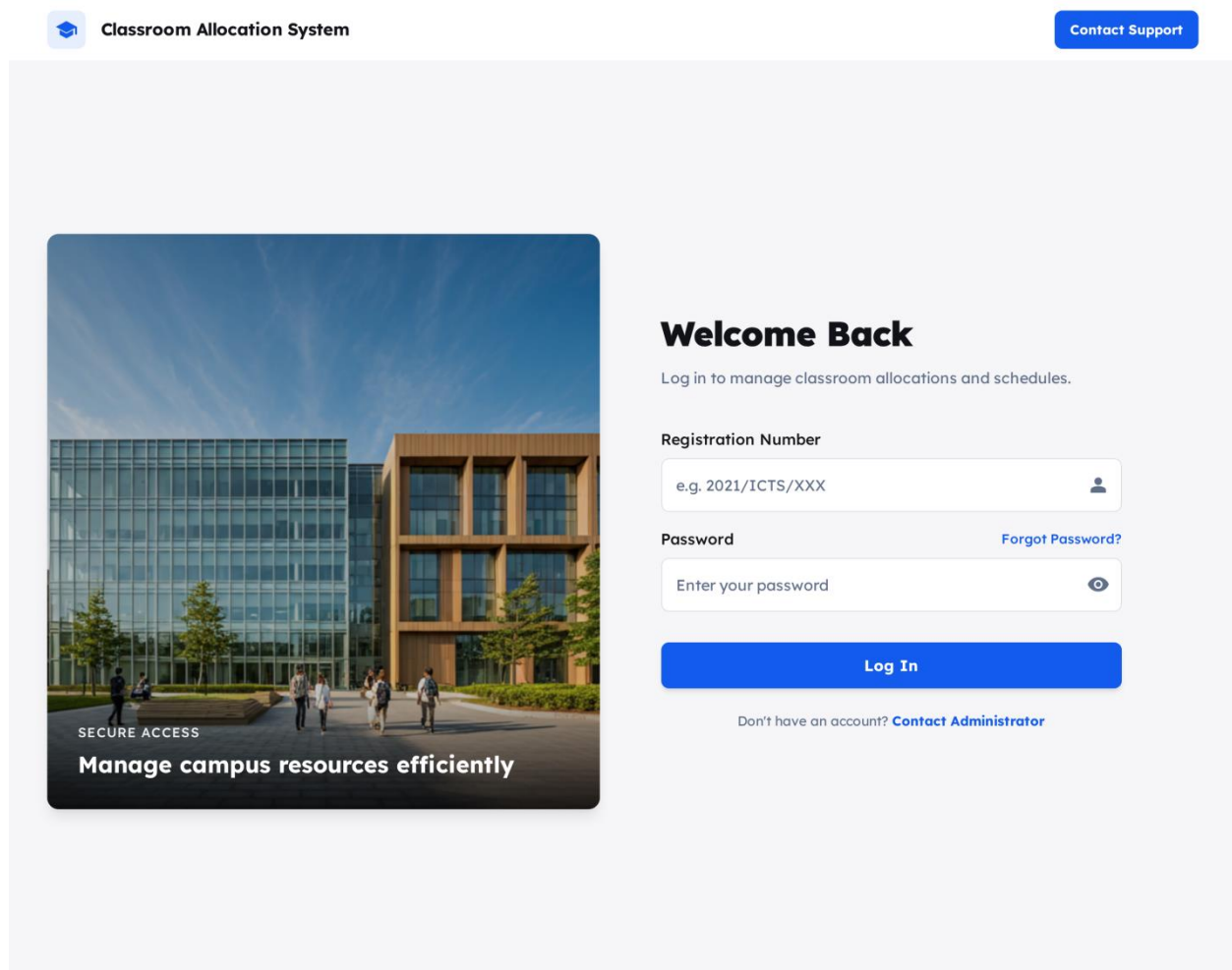
Backend: PHP

Database: MySQL

Server: Apache (XAMPP / WAMP)

9. Conclusion

The Classroom Allocation Management System provides an effective solution for managing classroom scheduling challenges. Automating the allocation process improves efficiency, accuracy, and overall institutional performance.



CAMS Admin

Dashboard

Classrooms

Schedule

Reports

Settings

James Admin

System Manager

Dashboard Overview

Search classrooms, teachers...

Good Morning, James

Here's what's happening in your campus today.

Export

New Allocation

Total Classrooms

45

→ Stable

Current Occupancy

78%

↗ +5% vs last week

Pending Requests

12

Action Needed

Maintenance Issues

4

Priority High

Weekly Utilization Trends

Average occupancy per day across all buildings

This Week

Mon

Tue

Wed

Thu

Fri

Sat

Quick Allocation

Instantly find and book an available room for now.

Book Now

ADMIN ACTIONS

Report Issue

Edit Sched

Daily Report

Teachers

Happening Now

Current sessions and immediate upcoming classes (Next 2 Hours)

View Full Schedule

TIME	COURSE CODE	ROOM	INSTRUCTOR	STATUS	ACTION
09:00 - 10:30	CS101 - Intro to CS	Lab A-201	Dr. Sarah L.	In Progress	
10:30 - 12:00	PHYS101 - Intro to Physics	Room 201	Prof. Mark D.	Upcoming	

EduAdmin

School Management System

Dashboard

Classrooms

Timetable

Faculty

Settings

Jane Administrator

Admin

Home / Administration / Classrooms

Classroom Management

Manage room allocation, capacity, and equipment details.

Add Classroom

Search by Room Name or Number

Building: All

Min Capacity

Features: Any

Status: All

	LECTURE HALL	BUILDING	CAPACITY	FEATURES	STATUS
☐	101 Room 101 Lab	Science Block	30	Projector Whiteboard	Active
☐	204 Room 204 Studio	Arts Wing	15	Easels Sink	Maintenance
☐	AUD Auditorium A Lecture Hall	Main Hall	200	Sound Stage	Active
☐	305 Room 305 Classroom	Science Block	35	Projector	Active

Showing 1 to 4 of 24 results

<

1

2

3

...

>

Calendar icon

Classroom Allocation

Fall 2024

Search for courses, rooms, or instructors...

🔔

⚙️

Alex Morgan
Registrar Admin

Allocations

UNASSIGNED 3 Pending

Lecture

CS101 - Intro to Algo

120 90m

Exam

PHY202 - Lab Exam

40 120m

Seminar

BIO305 - Adv. Genetics

25 60m

BUILDINGS

Engineering A

Floor 1 (Labs)

Floor 2 (Lecture Halls)

Floor 3 (Offices)

Science Complex B

Create New Schedule

Master Schedule

Engineering Building A

Week Day Month

Sep 23 - 29, 2024

Print Export

TIME	MON 23	TUE 24	WED 25	THU 26	FRI 27
08:00 AM		08:00 - 09:00 MAT201 - Linear Alg Rm 201 • Prof. Alan Exam			
09:00 AM	09:00 - 10:30 CS101 - Intro CS Rm 304 • Prof. Smith Lecture				09:00 - 10:30 CS101 - Intro CS Rm 304 • Prof. Smith Cancelled
10:00 AM			10:00 - 12:00 PHY202 - Physics II Lab 1 • Dr. Green Lab Requires Projector		
11:00 AM		Conflict			
12:00 PM				12:00 - 01:00 Guest Seminar Auditorium B Seminar	
01:00 PM					
02:00 PM					
03:00 PM					

Conflict Detected
Room 304 is already booked by ENG101 at 09:00 AM on Tuesday.
Find Alternative Dismiss

EduAllocate
Admin Console

Dashboard

Classrooms

Schedule

Reports

Settings

Home > Management > Reports

Reports & Analytics

Generate insights on classroom usage, conflicts, and efficiency to optimize campus resources.

Help Guide

Room Utilization

Scheduling Conflicts

Department Efficiency

Maintenance Logs

Report Filters

DATE RANGE

October 2023

29 30 1 2 3 4 5 6 7 8 9 10 11 12 13 14

Oct 5, 2 - Oct 11,

BUILDING

All Buildings

ROOM TYPE

Any Type

Generate Report

Utilization Report

Oct 5, 2023 - Oct 11, 2023 • All Buildings

PDF CSV Print

Avg. Occupancy

78%

+5% vs last week

Total Conflicts

3

Requires attention

Underutilized Rooms

12

Less than 40% occupancy

WEEKLY OCCUPANCY TRENDS

Mon Tue Wed Thu Fri

DETAILED RECORDS

Search records...

ROOM	TYPE	BUILDING	AVG. USAGE	STATUS
Room 101	Lecture Hall	Science Complex	92%	Optimal